

Agile enterprise architecture modelling: Evaluating the applicability and integration of six modelling standards



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ARTICLE INFO

Article history:

Received 8 November 2014

Received in revised form 25 April 2015

Accepted 7 July 2015

Available online 20 July 2015

Keywords:

ArchiMate

Agile modelling

Enterprise modelling

Enterprise architecture

Modelling standards

ABSTRACT

Context: Agile enterprise architecture artefacts are initially architected at the high-level and the details of those artefacts iteratively evolve in small project increments. There is a need to model agile enterprise architecture artefacts both at the high and low detailed level for a particular context. ArchiMate is relatively a new high-level architecture modelling standard. There is a growing interest amongst organisations in applying ArchiMate for high-level agile enterprise architecture modelling. However, organisations are unsure how to effectively apply ArchiMate at high-level and integrate it with their existing low detailed level modelling standards in practice for supporting end-to-end agile enterprise architecture modelling.

Objective: This paper evaluates the applicability and integration of high-level ArchiMate modelling standard with the existing low-level modelling standards such as BPMN (Business Process Model and Notation), UML (Unified Modelling Language), FAML (FAME [Framework for Agent-Oriented Method Engineering] Language), SoaML (Service Oriented Architecture Modelling Language), and BMM (Business Motivation Model).

Method: A qualitative questionnaire-based evaluation criteria has been developed based on the well-known and comprehensive The Open Group Architecture Framework (TOGAF). The evaluation criteria has been applied to evaluate the applicability and integration of the selected six modelling standards from the business, application, infrastructure and extension perspectives.

Results: Each modelling standard is different in scope. A single modelling standard usually does not provide the kind of support required by the agile enterprise architecture modelling. Based on the review results, a hybrid enterprise architecture modelling approach is proposed. This paper demonstrates the application of the proposed hybrid approach with the help of an agile enterprise architecture modelling case study.

Conclusion: It is concluded that the ArchiMate does not replace the existing low-level modelling standards, rather it can be used in conjunction with low-level modelling standards. This calls for the adoption of hybrid and integrated approach for agile enterprise architecture modelling.

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1. Introduction

Agile enterprises operate in a highly complex and dynamic environment. They require appropriate agile enterprise architecture (EA) capability for consistent and smooth enterprise operations and adaptation [1]. Indeed, continuous enterprise operations and adaptations are generally considered to be dependent on the provision of an agile EA capability [2,3]. The architecture of an agile enterprise can be defined as the “fundamental concepts or properties of a system in its environment embodied in its elements, relationships,

and in the principles of its design and evolution” [4]. Here, system refers to an agile enterprise. EA is a strategic discipline, which is critical for developing and realising an enterprise software strategy and roadmap [5]. The modern distributed and complex agile enterprise software supply chain, outsourcing and collaboration trends suggest the need for establishing a holistic agile EA capability [6].

Agile EA capability is important for proactively reviewing and assessing the operating environment and for seeking any changes or opportunities for improvement and transformation in the context of distributed and complex enterprise software supply chains [7,8]. The establishment of a situation-specific agile EA capability requires the adoption of a standard approach for modelling the agile EA artefacts at the high and low detailed level. There are

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number of high (e.g. ArchiMate, ADL) and low detailed level modelling standards (e.g. BPMN, UML). ArchiMate [9] is relatively a new high-level architecture modelling standard. There is a growing interest amongst organisations in applying ArchiMate for high-level agile enterprise architecture modelling [10]. However, organisations are unsure how to do so [11,12]. Research is scarce into how to effectively proceed with the adoption of enterprise modelling [13]. This draws our attention to the following research challenge?

How to effectively adopt ArchiMate at the high-level and integrate it with the existing low detailed level modelling standards in practice for supporting end-to-end agile enterprise architecture modelling?

This paper evaluates the applicability and integration of the high-level ArchiMate [9] with the five low detailed level modelling standards such as BPMN [14], UML [15], FAML [16], SoaML [17] and BMM [18]. This paper adopted a top (e.g. ArchiMate) to bottom (e.g. BPMN, UML) approach to selecting the six well-known modelling standards for review that fit in the context of research question in hand (see evaluation criteria and approach section for details). Further, it proposes the adoption of a situation-specific hybrid modelling approach for describing and analyzing agile EA artefacts as a possible precursor to establishing an overall modelling environment and tool support appropriate for comprehensive modelling of the agile EA artefacts. The application of the proposed hybrid modelling approach is demonstrated with the help of an agile EA modelling case study involving the ArchiMate for high-level modelling, and BPMN and UML (commonly used in industry) for low detailed level modelling.

This paper is organised as follows. Firstly, it discusses the research context. Secondly, it discusses the evaluation criteria and approach. Thirdly, it evaluates the applicability and integration of the six modelling standards: ArchiMate, BPMN, UML, FAML, SoaML and BMM. Fourthly, it discusses the evaluation results. Fifthly, based on the evaluation results, it proposes and demonstrates a hybrid and integrated modelling approach for supporting the agile EA capability. Finally, it concludes with key learnings and future research directions. We emphasise that the scope of this paper is to investigate the challenge of high-level ArchiMate adoption and integration with the existing low-level modelling standards in practice rather than providing a commentary and state-of-the-art on the applicability and integration of all the modelling standards. Of course, the evaluation approach presented in this paper can be applied to evaluate other modelling standards.

2. Research background and context

Agile EA does not suggest the detailed up-front architecture approach. Agile EA artefacts are initially modelled at the high-level. The detailed of the agile EA artefacts evolve through different agile project(s) increments. Agile EA capability requires the situation-specific practices, roles and tools for architecting architecture artefacts both at the high and low detailed level [19]. There are a number of EA frameworks such as that of Zachman [20], Federal Enterprise Architecture [21] and The Open Group Architecture Framework (TOGAF 9.1) [22] that can be used by organisations pursuing the establishment of an agile EA capability. These frameworks are different in their nature and scope and, therefore, a single framework may not be used as a universal standard for establishing an agile EA capability [19]. The elements from the EA frameworks can be combined to tailor a situation-specific agile EA capability. For instance, the Zachman [20] framework provides the EA ontology. TOGAF 9.1 [22] provides a concrete EA method. A situation-specific agile EA capability can be established by combining the EA ontology from Zachman [20] and EA method from TOGAF 9.1 [22]. This suggests that a situation-specific

adaptive or agile EA capability can be established by integrating the necessary elements from different well-known EA frameworks.

The establishment of a situation-specific agile EA capability also requires the adoption of a standard approach for modelling the agile EA artefacts. As discussed earlier, there are a number of EA modelling standards and commercial tools to choose from. However, a single EA modelling standard usually does not provide end-to-end (high to low level) modelling support as is required by the agile EA capability. ArchiMate [9] is relatively a new modelling standard. Organisations are showing strong interest in adopting ArchiMate. However, the question is how to effectively apply and integrate high-level ArchiMate with other low-level modelling standards for supporting end-to-end agile EA modelling both at the high and low detailed level. This marks the need for investigating the applicability of modelling standards and identify the opportunities for research and development. The scope of this paper is to review the applicability of six well-known modelling standards, and based on the review results, it propose a hybrid situation-specific agile EA modelling approach for describing and analyzing the agile EA artefacts both at the high and low detailed level for a particular context. There are at least over twenty modelling standards [23]. There are also a number of studies comparing similar standards such as comparing different agent [24] and service oriented modelling [25]. However, these studies do not report the cross comparison of modelling standards for high and low level modelling such as discussed in this paper. Unlike existing studies comparing similar standards, this paper is focused on how to effectively apply ArchiMate at high-level and integrate it with the existing low detailed level modelling standards in practice for supporting end-to-end agile enterprise architecture modelling. Therefore, the scope of this paper is limited to only six well-known modelling standards in practice (due to research scope and space limitations). However, similar review approach can be applied to other modelling standards.

3. Method: evaluation criteria and approach

TOGAF 9.1 [22] is a well-known comprehensive EA framework. It provides three main layers: business, application and technology or infrastructure layers. Additional layers or extensions can be added, if required. This paper uses the three layers from TOGAF and possible extension(s) for evaluating the applicability and integration of selected modelling standards (see Table 1). This paper adopted a top (high-level) to bottom (low detailed level) evaluation approach for reviewing the EA modelling standards (see Table 1). The top to bottom approach seems appropriate in the context of this paper as it allowed to review the modelling standards, from high to low level, as required and fit in the context of research question in hand. ArchiMate is selected and reviewed as an overarching high-level agile EA modelling standard. ArchiMate is organised into three layers (business, application, and technology). It also specifies two extensions (motivation, implementation and migration). ArchiMate is selected as a baseline high-level modelling standard to match with and review other low detailed-level modelling standards.

Other five modelling standards are selected as needed and required to support the ArchiMate for detailed-level agile EA modelling. For instance, BPMN supports business process modelling. BPMN is selected and reviewed from the ArchiMate business layer perspective. BMM supports strategy modelling. BMM is selected and reviewed from the ArchiMate motivation extension perspective. UML supports the object oriented application modelling. UML is selected and reviewed from the ArchiMate application layer perspective. FAML supports the agent oriented application modelling. FAML is selected and reviewed from the ArchiMate application layer perspective. SoaML supports the service oriented

Table 1
Review criteria.

Agile EA modelling	Modelling standards and elements					
	ArchiMate	BPMN	UML	FAML	SoaML	BMM
Business layer	How does ArchiMate support business layer for the high-level agile EA modelling?	How do other standards support and can be integrated with ArchiMate business layer for the low detailed level agile EA modelling?				
Application layer	How does ArchiMate support application layer for the high-level agile EA modelling?	How do other standards support and can be integrated with ArchiMate application layer for the low detailed level agile EA modelling?				
Technology layer	How does ArchiMate support technology layer for the high-level agile EA modelling?	How do other standards support and can be integrated with ArchiMate technology layer for the low detailed level agile EA modelling?				
Extension	Does ArchiMate offer any extension(s) for the high-level agile EA modelling?	How do other standards support and can be integrated with ArchiMate extension(s) for the agile EA modelling?				

application modelling. SoaML is selected and reviewed from the ArchiMate application layer perspective. In summary, UML, FAML and SoaML are paradigm specific (object, agent and service) software application modelling standards. These three different paradigm specific standards were selected for this review because they provide modelling support for well-known object, agent and service oriented paradigms. One may choose a different modelling standard for object or agent or service paradigm. Similar to this review paper approach, organisations may select a different high-level modelling standard as a baseline to review additional detailed-level modelling standards, if required. It is important to understand that the purpose of this review is to look at the practical applicability or use of these standards as opposed to their visual notation or syntax or structural comparison. The review criteria is summarised below.

4. Evaluation results: applicability and integration of modelling standards

This section applies the review criteria (see Table 1) and presents the review and applicability of six well-known modelling standards: ArchiMate, BPMN, UML, FAML, SoaML, and BMM (see Table 2). As discussed earlier, the purpose of the paper is to review these standards from application perspective as opposed to from visual syntax, semantics and structure perspective. The elements from these six modelling standards' metamodels and specifications have been reviewed, extracted and mapped in Table 2 by using the review criteria (see Table 1).

4.1. ArchiMate

ArchiMate [9] is relatively a new high-level architecture modelling standard. It provides a comprehensive EA modelling language that has been developed for formally describing and analyzing the architecture of an enterprise at the high-level. It provides a meta-model and graphical notation for visually describing and analyzing the three EA domains (business, application and technology architecture), their relationships and dependencies. ArchiMate provides three layers (business, application and technology) that can be used to describe the corresponding three EA domains and two extensions. Further, ArchiMate elements are also organised into active, behavioural and passive structures (see Table 2).

The business layer describes the business elements and their relationships within the business architecture domain. The business layer of the ArchiMate has been reviewed and the six key elements and their relationships have been identified: business objective, product, business process/function/interaction, business actor/role, business service and location (see Table 2).

The application layer describes the software application architecture elements and their relationships. The application layer of the ArchiMate has been reviewed and the four key elements and their relationships have been identified: data object, application service, application component and application interface (see Table 2). Application layer supports the business layer.

The technology or infrastructure layer describes the technology architecture domain elements and their relationships. The technology layer of the ArchiMate has been reviewed and the four key elements and their relationships have been identified: artefact, infrastructure service, node, and network/communication path (see Table 2). The technology layer supports the business and application layers.

ArchiMate includes the motivation extension, and implementation and migration extension. The motivation extension recognises the concepts of context or reasons behind the EA work. The implementation and migration extension recognises the concepts of portfolio, program, project (implementation), and plateau (migration).

There are a number of relationships that can be derived within and across layers from the elements embedded in these three layers and extensions. For instance, the business object in the business layer is realised by the data object in the application layer. Furthermore, the data object in the application layer is realised by the "artefact" in the technology layer, which could be a physical or executable file. Indeed, an artefact could be deployed on a node via an assignment relationship. In summary, ArchiMate provides an overarching high-level standard modelling language and graphical notation, which can be used to model agile EA artefacts at the high-level.

4.2. BPMN

BPMN [14] is a well-known business process modelling standard. It provides a standard business process model notation for formally describing and analyzing the business processes in detail. It provides five types of elements: flow, data, artefact, swimlane and connector for visually describing and analyzing business processes. Flow elements describe business process activities, tasks, events and gateways. Data elements describe business data objects and data stores. Artefact elements describe business process-related textual descriptions, annotations and groups (e.g. representing a grouping of elements). Swimlane elements describe business process orchestration and choreography patterns by using pool and lane elements. Connector elements describe the association, message flow and sequence flow between different elements. The scope of BPMN is strictly limited to the modelling of business processes, and the business strategy, organisational structure, functional breakdowns, data, information and rules models are considered to be out of scope. BPMN elements have been reviewed and mapped in Table 2 by using the review criteria.

4.3. UML

UML [15] is also a well-known software application modelling language. It was originated in the context of object oriented modelling. It provides a general purpose modelling language that has been developed under the auspices of the Object Management Group (OMG) for describing software applications in detail. OMG provides UML specifications in two parts: UML infrastructure and superstructure. UML infrastructure describes the foundational

Table 2
Evaluation results.

Agile EA modelling	Modelling standards and elements					
	ArchiMate	BPMN	UML	FAML	SoaML	BMM
Business layer	Active Structure: Business actor Business role Business collaboration Business interface Location	Pool Swimlane	Actor Swimlane Collaboration	Agent Role	Participant Agent	–
	Behavioral: Business process Business function Business interaction Business event Business service	Activity Task Gateway Sequence Events Pool Swimlane	Activity Use case Event Swimlane	Task	Service	–
	Passive structure: Business object Representation Meaning Value Product Contract	Data object Data store	Object	–	Service information Service contract	–
Application layer	Active structure: Application component Application collaboration Application interface	–	Class Component Collaboration Interface	Agent	Service interface Collaboration	–
	Behavioral: Application function Application interaction Application service	–	Method Interaction	–	Service	–
	Passive structure: Data object	–	Object	Message schema	Service information	–
Technology layer	Active structure: Node Device System software Infrastructure interface Network Communication path	–	Node	–	Service interface	–
	Behavioral: Infrastructure function Infrastructure service	–	Method	–	Service	–
	Passive structure: Artefact	–	Artefact	–	–	–
Extension	Motivational concepts: Stakeholder Driver Assessment Goal Requirement Constraint Principle	–	–	–	–	Mission Strategy Tactic Vision Goal Objective External influencer Internal Influencer Assessment Business policy Business rule
	Implementation and migration concepts: Work package Deliverable Plateau Gap	–	–	–	–	–

elements that are required to create the UML. UML superstructure describes UML from a usage and user perspective, which is aimed at creating actual model diagrams. The UML meta-model provides three types of key diagrams: behaviour, interaction, and structure diagrams for visually describing and analyzing the software applications. Behaviour diagrams describe the activity (including swimlane), use case (including actors) and event or state transitions. Interaction diagrams describe object collaboration and sequence. Structure diagrams describe object class, object, component (including interfaces and operations) and deployment nodes. The

scope of the UML is limited to the modelling of software applications such that business strategy, organisational structure, functional breakdowns and rules models are out of scope. Although business processes can be described by using UML activity diagrams; however, BPMN provides a much richer set of elements and greater depth and detail in the modelling of business processes. Therefore, it is not generally recommended to use UML as the sole or specialised modelling language for business processes. UML elements have been reviewed and mapped in Table 2 by using the review criteria.

4.4. FAML

FAML [16] is an emerging general purpose modelling language for describing agent-oriented software applications in detail. FAML provides support for both design-time and run-time agent oriented software application models. Here, the scope of the paper is to review the design-time models (e.g. architecture design); therefore, this paper does not include the discussion about the FAML run-time models. FAML design-time describes the agent-oriented software applications both at the individual agent (agent-internal) and agent system level (agent-external). An agent is an autonomous and interactive software component. An agent system is made of a multiple autonomous agents that scan, sense and respond to internal and external changes. The FAML agent-internal definition diagram describes an individual agent. The FAML agent-external system diagram describes an agent system. An agent system has agent definition, role, task, message schema, policy, ontology, and requirement elements. The scope of the FAML is limited to the modelling of agent-oriented software applications. Although, organisational structure can be described by using FAML agent (e.g. human, department) and role elements; however, ArchiMate provides a much richer set of elements and greater support and detail in the modelling of organisational structure. Therefore, it is not appropriate to use FAML as the modelling standard for organisational structure. UML can be used to model object oriented software application architecture in detail whereas FAML can be used to model agent oriented software application architecture in detail. Here, it is important to note that FAML does not provide specific visual modelling notation. One may extend and adopt specific notation from UML for modelling FAML elements. For instance, an agent can be modelled as a UML actor or class or component. FAML elements have been reviewed and mapped in Table 2 by using the review criteria.

4.5. SoaML

SoaML [17] is also an emerging general purpose modelling language that has been developed under the auspices of OMG. SoaML extends UML for describing service-oriented software application architecture in detail. The SoaML meta-model provides the support for modelling service, service provider interface, service consumer interface, service protocol, service information, service consumer, service provider, service policy, service organisation, service constraint and service usages. The scope of the SoaML is limited to the modelling of service-oriented software application architecture. Although, business services can be described by using SoaML service elements (e.g. human, department); however, ArchiMate provides a much richer set of elements and greater support and detail in the modelling of business services. Therefore, it is not generally recommended to use SoaML as the modelling standard for business services. UML can be used to model object oriented software application architecture in detail, FAML can be used to model agent oriented software application architecture in detail; and SoaML can be used to model service oriented software application architecture in detail. SoaML elements have been reviewed and mapped in Table 2 by using the review criteria.

4.6. BMM

BMM [18] is also a well-known business strategy or plan modelling language. It provides a general purpose modelling language that has been developed under the auspices of the Object Management Group (OMG) for describing business plan elements. BMM has two parts. The first part specifies the means and ends of the business plan. Ends are strategic business goals and objectives.

Means refer to actions such as strategies, tactics, policies and rules that an organisation can employ to achieve the ends. The second part includes the elements that influence the means and ends such as the assessment of strengths, weaknesses, opportunities, and threats. ArchiMate specifies motivation extension concepts; however, BMM provides a much richer set of elements and greater support and detail in the modelling of business motivation. Hence, it is appropriate to model motivation using the BMM; and portfolio, program, project and plateau using the ArchiMate implementation and migration extension. BMM elements have been reviewed and mapped in Table 2 by using the review criteria.

5. Discussion and analysis

ArchiMate is a comprehensive high-level EA modelling standard. Therefore, ArchiMate elements and layers are used as a baseline and starting point for this review. Table 2, column 1, outlines the review criteria: business layer, application layer, technology layer and extension. Column 2 maps the elements from ArchiMate to each agile EA layer and extension. Column 3 maps the BPMN elements to each layer corresponding to ArchiMate layer. Column 4 maps the UML elements to each layer corresponding to ArchiMate layer. Column 5 maps the FAML elements to each layer corresponding to ArchiMate layer. Column 6 maps the SoaML elements to each layer corresponding to ArchiMate layer. Column 7 maps the BMM elements to corresponding ArchiMate extension layer. If any of the modelling standards such as BPMN or UML do not offer the elements corresponding to ArchiMate layer then “–” symbol is placed in the relevant cells (see Table 2). The results of the review indicate that ArchiMate provides an overarching comprehensive set of modelling elements that can support all three layers and possible extensions of agile EA at high-level (see Table 2).

ArchiMate is appropriate for the high-level modelling of the business processes within the overall business architecture context. However, detailed business process modelling requires much richer elements. Therefore, once the business processes are identified and modelled at the high-level using the ArchiMate, then later BPMN can be used for detailed business process modelling (just-in-time modelling) as the more information about the processes is emerged during agile projects. The scope of BPMN is strictly limited to the modelling of business processes of the business architecture domain. BPMN only supports some elements for the business layer in comparison to ArchiMate. BPMN also does not support the application and technology layers. This means that other aspects of the agile EA, which are not in the scope of BPMN, perhaps, can be modelled by using the ArchiMate. This analysis also draws our attention to the possible integration of ArchiMate and BPMN at the business process level. Since, ArchiMate can be used for initial high-level business process modelling. Later, as the business architecture details are evolved then such business process details can be modelled just-in-time using the BPMN. It is not desirable in agile EA to detail the business processes up-front. The combination of ArchiMate and BPMN seems appropriate for business architecture modelling in the overall context of agile EA modelling.

Secondly, it is also noted in the ArchiMate specifications that it is appropriate for identifying the application elements, interfaces, usages and their modelling at the highest level. However, the detailed internal structure, interactions and behaviour of an individual application is intentionally beyond the scope of ArchiMate, which can be described by using the UML (object oriented application modelling), FAML (agent oriented application modelling) and SoaML (service oriented application modelling). Further, it can be observed from the analysis results (see Table 2)

that, although UML supports application modelling in detail, some of the UML elements can also be used for describing some of the ArchiMate business architecture layer elements. Similar to BPMN, UML can be used for detailed business process modelling by using a combination of business activity and business use case diagrams. However, UML is not a specialised and rich language for modelling the business processes when it is compared to BPMN. Therefore, it is appropriate to use BPMN for describing the business processes identified in the business architecture of an agile EA. Anyhow, once the applications and their interactions are identified and modelled at the high-level using the ArchiMate, then later UML or FAML or SoaML can be used for detailed individual application architecture modelling (just-in-time modelling) as the more information about the application is emerged during agile projects. It is not desirable in agile EA to detail the individual application architecture up-front. The combination of ArchiMate, UML, FAML and SoaML seems appropriate for application architecture modelling in the overall context of agile EA modelling.

It is important to note here that application portfolio of an organisation could have a number of software applications and agile projects. These applications or projects could be object oriented, agent oriented and service oriented. If all or some components of a software application are object oriented abstraction specific then UML is a specialised standard for modelling object oriented paradigm specific software applications (or parts of the applications) in detail (e.g. UML behaviour, interaction and structure diagrams); therefore, it seems to be appropriate to use UML in combination with ArchiMate at the application architecture layer level. If all or some components of a software application are agent oriented abstraction specific then FAML (e.g. agent, agent system) can be used to model the agent oriented components of the software application in detail (see Table 2). It is important to note here that if all or some components of a software application are service oriented abstraction specific then SoaML (e.g. service, service interface) can be used to model the service oriented components of the software application in detail.

It is clear from the analysis that individual software “Application” modelling is the possible integration point between ArchiMate application architecture (high-level modelling), and UML, FAML and SoaML (individual application low detailed level modelling). Furthermore, it can also be observed from the analysis (see Table 2) that ArchiMate provides a richer set of elements for modelling the technology or infrastructure architecture in detail in comparison to UML, FAML, and SoaML. Therefore, ArchiMate is the better option for modelling technology or infrastructure architecture artefacts. Finally, in addition to standard business, application and technology layers of an agile EA, it can also be observed from the analysis (see Table 2) that it is appropriate to model motivation using the BMM; and portfolio, program, project and plateau using the ArchiMate implementation and migration extension. BMM provides richer set of agile EA strategy or motivation modelling as compared to ArchiMate.

In summary, the results of this evaluation indicate that all six modelling standards have their scope and strengths, and a single standard usually cannot be adopted for fully supporting the agile EA modelling needs. These six standards complement the scope of each other. This analysis suggests the need for tailoring a hybrid EA modelling approach, which can be supported through available or new modelling tools. The hybrid EA modelling approach is discussed in the following section.

6. Hybrid EA modelling

Based on the review, this section proposes the hybrid EA modelling (HEAM) approach to support the agile EA capability (see Fig. 1). The HEAM proposes the use of ArchiMate as an overarching and specialised standard for modelling the agile EA artefacts at the high-level. This is because it provides the overall modelling coverage for all the three layers (e.g. business, application and technology architectures) of an agile EA and related necessary extensions. However, each business process from the business architecture layer can be better detailed using the BPMN. Business process

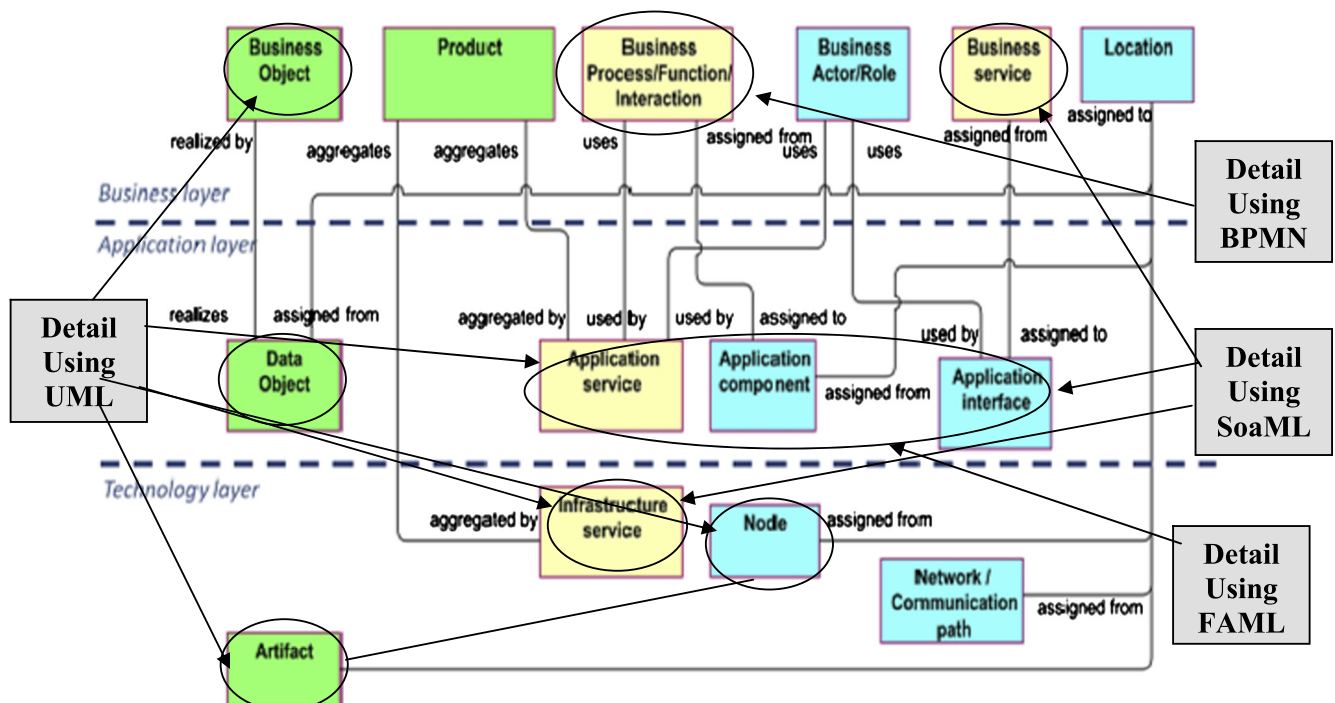


Fig. 1. ArchiMate core metamodel and mapping with BPMN, UML, SoaML, and FAML.

modelling is an integration point between the ArchiMate and BPMN as shown in Fig. 1 – see arrow and circle shapes. Furthermore, based on the analysis, it is clear that abstraction specific (e.g. object, agent, and service) individual application architecture can be better detailed by using UML, FAML and SoaML. In addition, application modelling is an integration point between ArchiMate, and UML, FAML and SoaML as shown in Fig. 1 – see the arrow and circle shapes. HEAM approach is further discussed at each layer of an agile EA in the following sections. Please note that ArchiMate core metamodel in Fig. 1 is copyrighted material from the ArchiMate 2.1 Specification [9] and is used with permission of The Open Group. ArchiMate is a registered trademark of The Open Group.

6.1. Business architecture: ArchiMate and BPMN combination

The business architecture domain of the agile EA can be modelled using the ArchiMate and BPMN. It is clear from the analysis that ArchiMate is a high-level language and is not designed for detailed business process modelling. However, ArchiMate can be used to model the business processes (including actors/roles) and interactions at the uppermost level in the overall business architecture and then one can use business process modelling elements from BPMN for modelling each business process details just-in-time (including actors/roles in process swimlanes and pools). Furthermore, ArchiMate can be used to model those elements that are not within the scope of BPMN such as business products and services, which includes value and contract, organisational charts (e.g. actors and roles), functions (e.g. functional decomposition), business object model, etc.

6.2. Application architecture: ArchiMate, UML, FAML and SoaML combination

The application architecture domain of the agile EA can be modelled using the ArchiMate and elements from UML, FAML and SoaML. It is clear from the analysis that the ArchiMate is not intended for detailed individual software application modelling. Therefore, ArchiMate can be supported through software application modelling elements from UML, FAML and SoaML for modelling software application details just-in-time. For instance, ArchiMate can be used to model the applications and their interactions at the highest level in the overall application architecture after which, depending on the application abstraction mechanism (object, agent and service), application modelling elements from the UML or FAML or SoaML can be used for modelling each application architecture in detail. In addition, ArchiMate can be used to model application usage (e.g. application interface, application collaboration) and application service, neither of which are appropriate to be modelled using the UML or FAML or SoaML.

6.3. Technology architecture: ArchiMate, UML and SoaML combination

The technology architecture domain of the agile EA can be modelled using the ArchiMate and elements from UML and SoaML. For instance, ArchiMate can be used to model the infrastructure service, node, artefact, network, communication path and their interactions at the highest level in the overall technology architecture after which, infrastructure service can be detailed using the UML “Method” or SoaML “Service” elements. Although UML and SoaML elements can be used, however, ArchiMate technology layer elements as a whole seem more appropriate for technology architecture modelling, as compared to other modelling standards such as BPMN, UML, FAML and SoaML. Therefore, ArchiMate can be used for modelling the entire technology architecture layer of the EA.

6.4. Extension: ArchiMate and BMM combination

The core metamodel of ArchiMate (Fig. 1) does not show the extension. It is clear from the analysis (see Table 1) that BMM is more appropriate for detailed motivation modelling (ArchiMate extension) when compared to ArchiMate. However, ArchiMate should be used for implementation and migration concepts since there is no such support for this in BMM or other standards reviewed in this paper. Overall, BMM is a specialised language for modelling motivational concepts and is not intended to be used for other layers and extension of the agile EA modelling.

In summary, business architecture domain of the agile EA can be modelled using the combination of ArchiMate and BPMN both at the high and low detailed level. Application architecture domain of the EA can be modelled using the combination of ArchiMate, UML, FAML and SoaML both at the high and low detailed level. Technology architecture domain of the EA can be modelled using the combination of ArchiMate, UML and SoaML both at the high and low detailed level. Finally, the extensions of agile EA (other than traditional business, application and technology domains) can be modelled using the combination of ArchiMate and BMM. Additional, modelling standards and combinations can be investigated using the approach presented in this paper. One may begin with the ArchiMate or any other similar high-level modelling standard and then can incorporate and review other low detailed level modelling standards as required. This is a top down approach. However, if an organisation is already using other standards such as BPMN and UML. then they may identify and adopt the parts of ArchiMate that are not well supported in their existing adopted standards (bottom-up approach). This review provides a knowledge base and review approach for practitioners and researchers who are interested in agile EA modelling.

7. Agile EA modelling case study

The application of the HEAM approach is demonstrated here with the help of an agile EA modelling case study example involving ArchiMate, BPMN and UML. The agile EA has three domains: business architecture, application architecture and infrastructure architecture.

7.1. Business architecture

Business architecture can be modelled or described in terms of a number of views such as organisation view, business capability view and business process view. An organisation can have a number of business capabilities as shown in Fig. 2. These capabilities are modelled or documented in a business capability map by using the ArchiMate. ArchiMate provides the business function element and notation to model the business capabilities. A simple business capability map is developed and presented (see Fig. 2) here to demonstrate the use of ArchiMate – business layer.

A business capability [26] defines what does a business do? A business capability can be realised by a one or many business processes. Once the business capability map is developed, then the next step is to identify and model the business capability processes. Here, one of the capability is “Procurement Management”. It has four processes: Submit Purchase Order Request, Place Order, Receive Products, and Make Payment. These four business processes realise the procurement management capability. Staff can submit a purchase order request to internal central procurement management capability. Procurement management capability processes the order through the engagement of an external supplier. ArchiMate can be used to model the procurement management business processes and their interactions at the

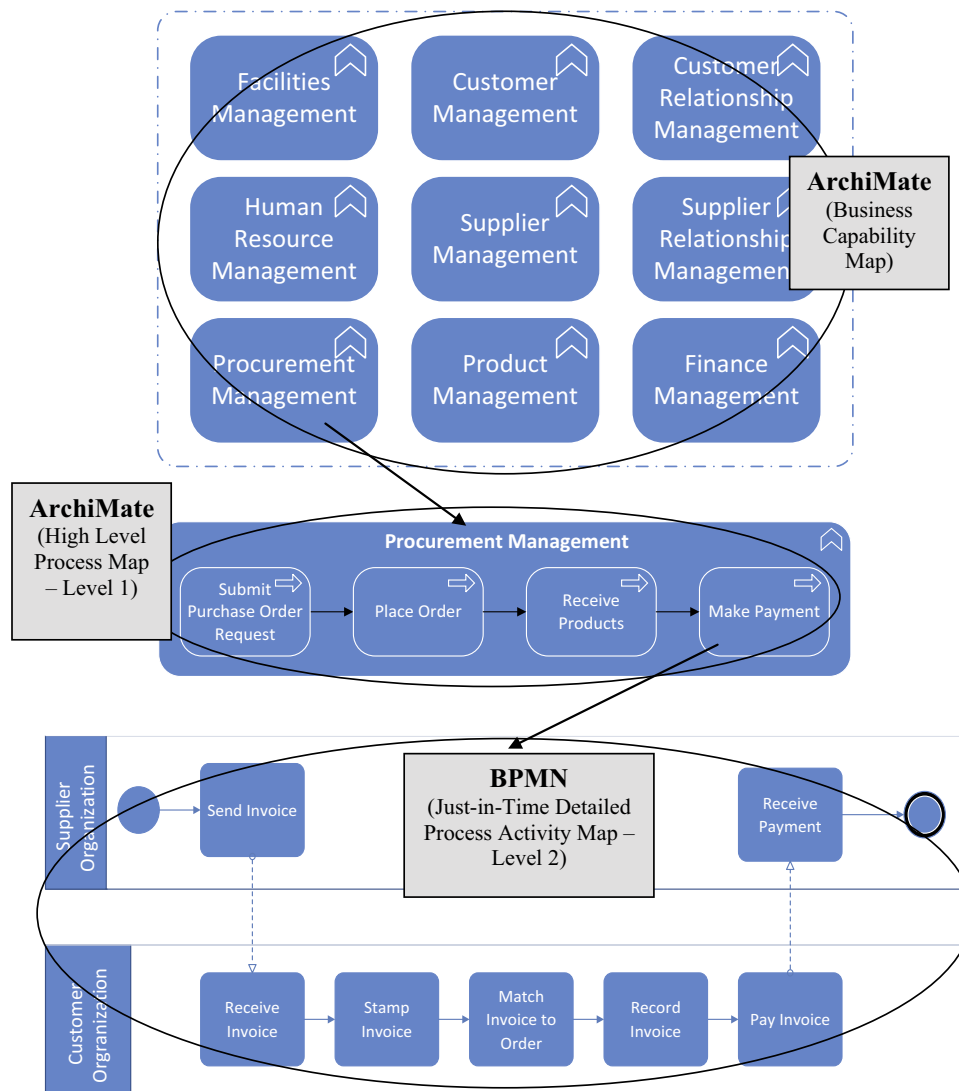


Fig. 2. Business architecture modelling: business capability map, high and low level process maps.

high-level (level 1 process map) in the overall business architecture (see Fig. 2). Further, each process can be detailed just-in-time by using the BPMN. Here, the “Make Payment” business process of the procurement management capability is modelled using the BPMN.

As noticed earlier that the ArchiMate is appropriate to model the business processes and their flow at the high-level (level 1 process modelling). However, BPMN is a specialised business process modelling language and provides a richer set of elements (e.g. pools, swimlanes, events, activities, etc.) to detail the each business process activities (level 2 process modelling). Hence, it can be observed here that how a business architect can use ArchiMate for modelling the business architecture artefacts both at the high (e.g. business capability map and business process model at level 1) and low detailed level (e.g. business process activities and interactions at level 2) for a particular context. Business process details can be further analyzed for identifying the user stories or requirements for agile projects [27].

7.2. Application architecture

Application architecture supports the business architecture. Application architecture shows (see Fig. 3) three applications

(order processing application, product management application and financial application) that support the procurement management business processes. ArchiMate provides the elements and notation that can be used to model the applications (e.g. application service, interface and component) and their interactions at a high-level in the overall application architecture. Order processing application, in the overall context of application architecture, is a component that realises the application services (e.g. purchase request service, order submission service) to support the business processes. The realisation relationship is modelled using the large arrow head shape (as per ArchiMate standard) in Fig. 3. Similarly product management application and financial application are components of the application architecture that realise product information and payment services respectively. These services can be exposed and accessed via application interfaces. Application interaction aggregates a number of applications for performing a particular task or transaction. Application interaction describes the application collaboration. The aggregation and describe relationships are modelled using the aggregation arrow head shape and round arrow head shape (as per ArchiMate standard) in Fig. 3. Each application architecture (e.g. software architecture) can be further detailed just-in-time by using the UML elements for a particular context or project. For instance,

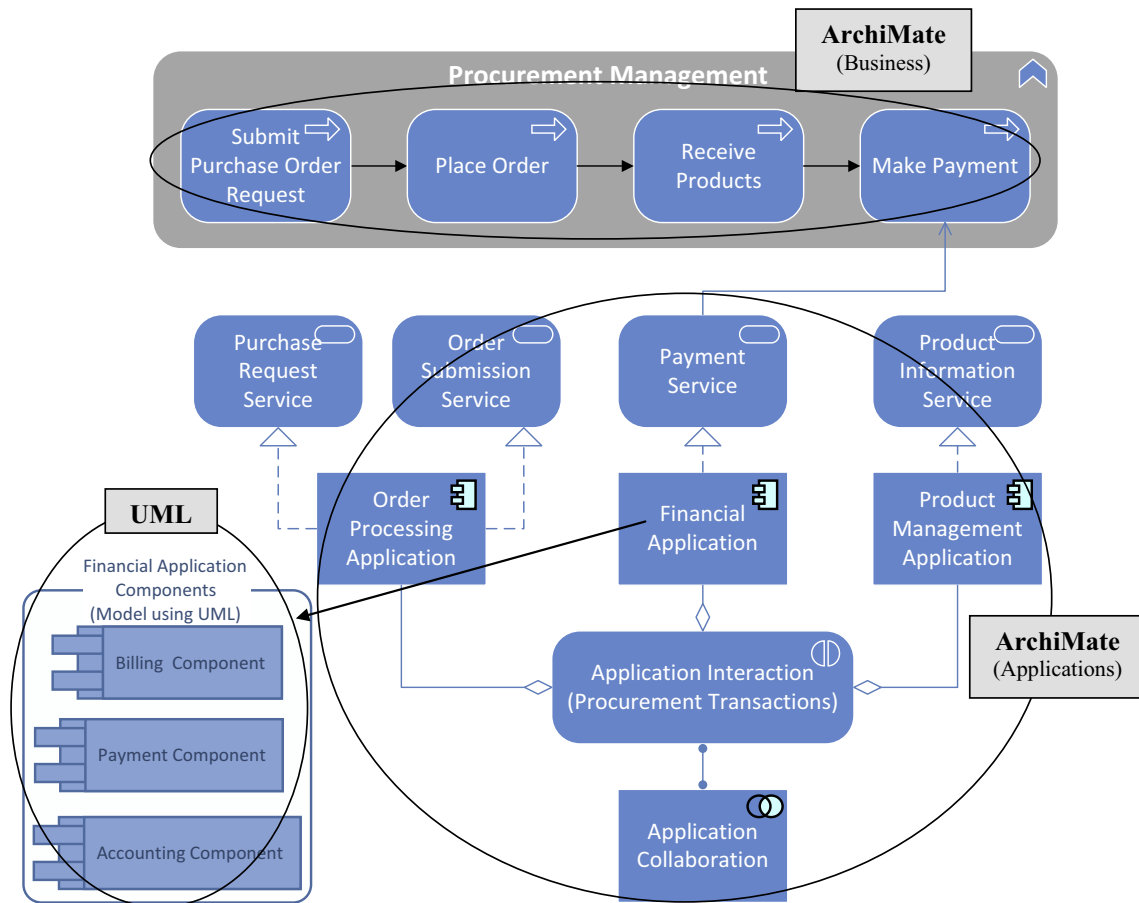


Fig. 3. Application architecture modelling.

the application architecture components and their interactions can be modelled using the UML component elements (only components are shown in UML box for simplicity reasons). Application service details (such as service operations) can be modelled using the UML interface class and method elements. Hence, a hybrid approach by combining ArchiMate and UML seems useful to model the application architecture at the high and low detailed level. Further, the application architecture can be guided thorough application architecture design patterns [28] such as model-view-controller (MVC), client-server architecture, pipe-and-filter, and repository architecture.

7.3. Infrastructure architecture

ArchiMate provides richer elements than BPMN and UML for infrastructure modelling. ArchiMate is used to demonstrate the modelling of infrastructure architecture that is required to support the procurement management application such as order processing, financial and product management applications. Here, a 3-tier layered infrastructure architecture pattern (e.g. application client tier, application server tier and database server tier) is used for the order processing, product management and financial applications. These application clients and servers can be modelled using the ArchiMate node and communication path elements. Node element includes system software and device elements to model the application client, application and database servers. The application client, application server and database server are connected via ArchiMate communication path element (see Fig. 4). Fig. 4, also shows how infrastructure architecture can support the application architecture and application architecture (e.g.

financial application) can support the business architecture. For instance, “Make Payment” business process (business architecture) uses the “Payment Service” (application architecture). “Payment Service” is realised by the “Financial Application” (application architecture). The infrastructure architecture components support the “Financial Application”. This also confirms that ArchiMate provides a support for high-level modelling for all three layers of the agile EA.

This section demonstrated the hybrid modelling approach with the help of an agile EA modelling case study. It seems that ArchiMate does not replace the existing well-known modelling standards. Rather, it can be used as an overarching standard in conjunction with detailed level modelling standards for a specific context. It is also important to note that agile EA modelling suggests using the informal modelling tools such as flip charts, white boards and sticky notes for modelling the agile architecture artefacts [29]. However, the artefacts modelled using the informal tools can be converted into formal artefacts using the formal but simple tools. The agile EA modelling case study described in this paper was done by using the available existing modelling tool (e.g. Visio). Visio already supports the UML, ArchiMate and BPMN stencils were imported into Visio to create a simple integrated hybrid modelling environment (see Fig. 5). Hence, it is not required to buy different tools for each different modelling standard. In the beginning, it is suggested to keep using your existing tool(s) and import relevant stencils (similar to this case study). However, depending on the maturity, size and requirements of the agile EA capability, one may consider adopting more sophisticated agile EA modelling and management tools such as Abacus, Systems Architect, and Orbus. This research does not endorse any specific

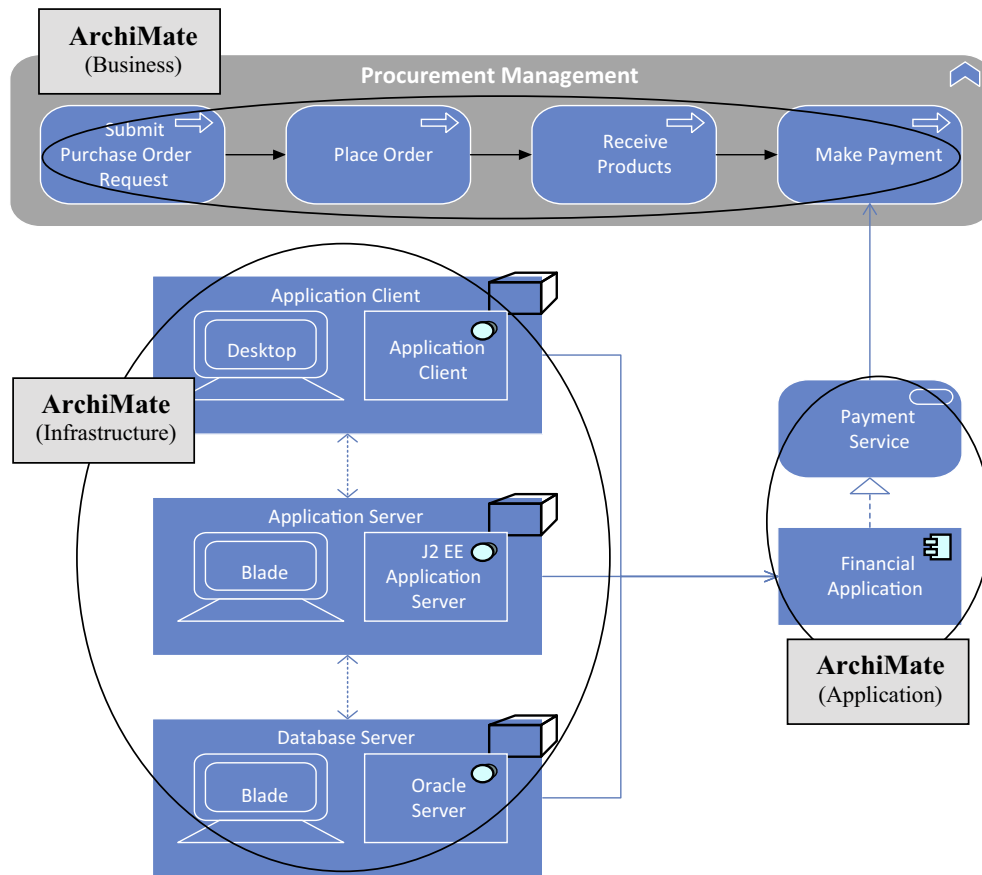


Fig. 4. Infrastructure architecture modelling.

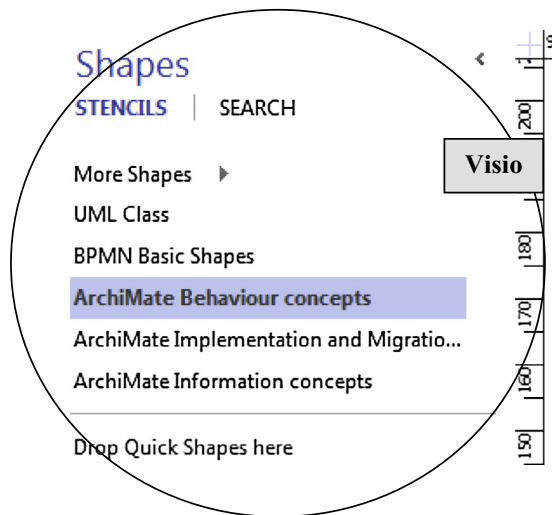


Fig. 5. Visio – simple integrated hybrid modelling environment.

tools, organisation should make their own assessment when choosing the tools for their agile EA capability. Further, this paper only offered a single case study. It is very challenging to present an end-to-end coherent case study involving all the enterprise architecture layers. Nevertheless, a single case study may be considered as a limitation of the work presented in this paper. The results presented in this paper may not be generalised, however, this paper can be used as a baseline for future studies in this important area of research.

8. Conclusion

This paper reviewed the six well-known modelling standards and proposed an integrated HEAM approach. The HEAM approach seems appropriate for modelling both the agile and non-agile EA artefacts both at a high and a low, fully detailed level. In contrast to non-agile EA modelling, in agile EA modelling, EA artefacts are iteratively evolved and therefore, they are not modelled in detail upfront. Agile EA is first modelled at the high level and low level details are incrementally emerged and modelled. The work presented in this paper has both research and practical implications. It will help agile EA researchers and practitioners to understand the scope and applicability of each modelling standard. The review and application of the HEAM approach may help practitioners to make an informed decision about the adoption, tailoring and integration of different modelling standards and tools for their local context. In our further research, we are aiming to develop an automatic model transformation engine that can automatically transform and link between ArchiMate, BPM, UML, FAML, and SoAML models. Automatic transformations require precise mappings and transformation between concepts in these metamodels; gaining an understanding of the detailed semantics, syntax and structure of these concepts, in ontological terms [30], and requires considerable effort.

Acknowledgements

I wish to extend my sincere thanks to architects from research and commercial organisations who helped with their valuable feedback and suggestions.

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